Homework 3

BSTA 513/613

Your name here - update this!!!!

> **Caution**
>
> Homework is ready to go!! (5/5/2025)

Purpose

This homework is designed to help you practice the following important skills and knowledge that we covered in Lesson 10-12:

- A bunch of work on interactions
- Identifying numerical problems and thinking through potential fixes
- Assessing overall measure of model fit

Directions

- Download the `.qmd` file here.
- You will need to download the datasets from our shared folder.
- Please upload your homework to Sakai. Upload both your .qmd code file and the rendered .html file
  - Please rename you homework as Lastname_Firstinitial_HW02.qmd. This will help organize the homeworks when the TAs grade them.
- For each question, make sure to include all code and resulting output in the html file to support your answers
- Show the work of your calculations using R code within a code chunk. Make sure that both your code and output are visible in the rendered html file. This is the default setting.

> **Tip**
>
> It is a good idea to try rendering your document from time to time as you go along! Note that rendering automatically saves your qmd file and rendering frequently helps you catch your errors more quickly.

Questions Part 1

Question 1

This question is taken from the Hosmer and Lemeshow textbook. The ICU study data set consists of a sample of 200 subjects who were part of a much larger study on survival of patients following admission to an adult intensive care unit (ICU). The dataset should be available in our shared folder. The major goal of this study was to develop a logistic regression model to predict the probability of survival to hospital discharge of these patients. In this question, the primary outcome variable is vital (survival) status at hospital discharge, STA. Clinicians associated with the study felt that a key determinant of survival was the patient's age at admission, AGE. We will build to a multivariable logistic regression model while adjusting for cancer part of the present problem (CAN), CPR prior to ICU admission (CPR), infection probable at ICU admission (INF), and level of consciousness at ICU admission (LOC).

A code sheet for the variables to be considered is displayed in Table 1.5 below (from the Hosmer and Lemeshow textbook, pg. 23). We refer to this data set as the ICU data.

You will need to use some of the mutations implemented in HW 2, Q2, Part d.

Table 1.5 Code Sheet for the Variables in the ICU Data

Variable	Description	Codes/Values	Name
1	Identification code	ID number	ID
2	Vital status at hospital discharge	0 = Lived 1 = Died	STA
3	Age	Years	AGE
4	Gender	0 = Male 1 = Female	GENDER
5	Race	1 = White 2 = Black 3 = Other	RACE
6	Service at ICU admission	0 = Medical 1 = Surgical	SER
7	Cancer part of present problem	0 = No 1 = Yes	CAN
8	History of chronic renal failure	0 = No 1 = Yes	CRN
9	Infection probable at ICU admission	0 = No 1 = Yes	INF
10	CPR prior to ICU admission	0 = No 1 = Yes	CPR
11	Systolic blood pressure at ICU admission	mm Hg	SYS
12	Heart rate at ICU admission	Beats/min	HRA
13	Previous admission to an ICU within 6 months	0 = No 1 = Yes	PRE
14	Type of admission	0 = Elective 1 = Emergency	TYPE
15	Long bone, multiple, neck, single area, or hip fracture	0 = No 1 = Yes	FRA
16	PO_2 from initial blood gases	0 = >60 1 = ≤ 60	PO2
17	PH from initial blood gases	0 = ≥ 7.25 1 = <7.25	PH
18	PCO_2 from initial blood gases	0 = ≤ 45 1 = >45	PCO
19	Bicarbonate from initial blood gases	0 = ≥ 18 1 = <18	BIC
20	Creatinine from initial blood gases	0 = ≤ 2.0 1 = >2.0	CRE
21	Level of consciousness at ICU admission	0 = No coma or deep stupor 1 = Deep stupor 2 = Coma	LOC

Part a

Write down the population equation for the logistic regression model of STA on AGE, CAN, CPR, and INF. How many parameters does this model contain?

Part b

Using `glm()`, obtain the maximum likelihood estimates of the parameters of the logistic regression model in Part a. Using these estimates, write down the equation with the fitted values.

Part c

Assess the significance of the group of coefficients for all variables in the model using the likelihood ratio test. (Hint: part of the ratio in the LRT will be an intercept only model)

Part d

Fit a new model using only CAN and INF as the predictors, including an interaction between CAN and INF. Is there evidence that our model should have an interaction between CAN and INF (Hint: this requires a formal test of the interaction)?

Part e

Interpret the odds ratio for the **main effects** in the model from Part d. Please include the 95% confidence interval.

Part f

From the above model, fill out the following table for the odds ratios. Note, you will only need to report two odds ratios and you already have one from Part d & e.

Cancer	Infection	Estimated odds ratio	95% CI
Cancer part of present problem	Infection probable at ICU intake		
	No		
	Yes	FILL HERE	FILL HERE
Cancer not part of present problem	Infection probable at ICU intake		
	No		
	Yes	FILL HERE	FILL HERE

This is a really good way to report odds ratios for interactions between two categorical predictors! Might want to keep this in mind for your project!!

Part g

Interpret the odds ratio from the table in Part e. Please include the 95% confidence interval. What do you notice about the odds ratios (Hint: Think back to my last slides in Lesson 10: Interactions)?

Part h

Compute the predicted probability for a subject who does not have a present issue with cancer nor an infection upon admittance to the ICU. Compute the 95% confidence interval for the predicted probability. Can you use the Normal approximation?

Part i

Interpret the predicted probability from Part h (right above), including the confidence interval.

Part j

Building off of Part e, fill out the following table for predicted probabilities. What do you notice about the predicted probabilities (Hint: Think back to my last slides in Lesson 10: Interactions)?

Cancer	Infection	Predicted probability	95% CI
Cancer part of present problem	Infection probable at ICU intake		
	No	FILL HERE	FILL HERE
	Yes	FILL HERE	FILL HERE
Cancer not part of present problem	Infection probable at ICU intake		
	No	FILL HERE	FILL HERE
	Yes	FILL HERE	FILL HERE